

Tidy a Bookshelf SOP (1x Shelf, in situ)

One episode tidies one shelf of a real bookcase, in place.

The base is **passive**. It has no drive of its own. It is pushed by hand to the bookcase and locked there. Everything the episode touches is already on the bookcase or on the side table beside it when recording starts.

The bookcase is worked **as found**. The **working shelf** is bare, with dust on its board and along its lip. Loose books lie flat in stacks on the **lower shelf**, the shelf directly below it, with non-books among them. The bin and the bookend stand on the **side table**. The episode ends with every book standing in one row on the working shelf, tallest to shortest, spines facing out, held up by the bookend. The non-books are in the bin. The used cloth lies on the side table. The shelf board and lip are clean.

This is an in-situ task. Two rules follow from that.

First, there is a shelf directly above each shelf being worked. So every approach is from the front, straight in through the shelf opening. No gripper ever comes down onto a shelf board from above.

Second, a book leaves the lower shelf only one way. It is gripped by the part of its spine that sticks out past the lower lip, drawn straight out to the front, level, handed over in front of the working shelf, and carried straight into the working shelf standing up. No book is lifted toward the shelf above it or carried over another book.

The order never changes. Wipe the shelf board and lip and put the cloth on the side table. Clear the non-books into the bin. Shelve the books, tallest first. Set the bookend last. The shelf is wiped before anything else is touched.

The two arms never cross. The left arm works the left. The right arm works the right. The left gripper always stays left of the right gripper. Things change grippers only by a hand-over in front of the shelf midline.

The bookcase is set up in one of three ways. Only the cloth moves: it starts at the left end, the middle, or the right end of the lower shelf. Everything else stands in the same place in both.

- **Config L:** the cloth starts at the **left** end of the lower shelf.
- **Config M:** the cloth starts in the **middle** of the lower shelf, at the shelf midline.
- **Config R:** the cloth starts at the **right** end of the lower shelf.

Where a step depends on the setup it says so on an **IF** line: look at the bookcase and follow the line that matches.

What stays constant across all sessions:

- **Start position:** one config per episode, chosen before recording and never changed mid-episode.
- **Row:** the row is always built at the **right end panel** and grows to the left. The non-books and the side table are always on the **left**.
- **Book rule:** the **left gripper** picks each book off the lower shelf and hands it over. The **right gripper** takes it, stands it up, and shelves it. When it is needed, the **right gripper** may first slide a book along the lower shelf so the left gripper can take it.
- **Left-side work:** the **left gripper** also clears the non-books into the bin, lays the used cloth on the side table, and hands the bookend over. The **right gripper** sets the bookend on the shelf, because the left arm cannot reach the end of the row.
- **Cloth rule:** the gripper on the cloth's side takes the cloth and wipes its own half first, then hands the cloth over at the shelf midline and the gripper on the far side wipes the other half. The **left gripper** starts in Config L. The **right gripper** starts in Config M and Config R.
- **Order:** the order of the work never changes with the config. Only which gripper takes the cloth first changes.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the bookcase. Stop it **square to the front of the bookcase**, so the shelf lip runs straight across the frame and neither end of the shelf sits nearer than the other.
2. Stop it **centered on the working shelf**, so the two end panels are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach the back of the working shelf and the back of the lower shelf straight in from the front without either arm extending. Stop it **far enough** that no arm, wrist, or part of the base touches the bookcase or the side table when both arms work at once.
4. Check the **shelf heights**. With the base parked, both grippers must enter both shelf openings level, reach the board and the back of each shelf, and travel the whole span

without the wrist or forearm touching the shelf above or a lip.

5. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
6. If any of lines 1 to 4 fails, push the base to a new park by hand and start again at line 1. Do not work shelves the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the furniture hard enough to shift it. No operator touches it. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the working shelf. Its frame includes the whole working shelf from end panel to end panel, the whole shelf lip, the whole lower shelf, the hand-over place, and the side table with the bin and the bookend on it.
3. The camera shows the shelf board and the shelf lip. You can see the dust on them and whether it has been wiped off.
4. The camera shows each book from the front. You can see its spine, whether it lies flat or stands, and how long it is against its neighbours.
5. Both arms are at home with grippers open.
6. The bookcase is steady. It does not rock when a book is pushed along a shelf board.
7. The right arm reaches the hand-over place, the build place, and the strip of working shelf board from the build place to the right end panel, straight in from the front and without extending to a joint limit. It also reaches the open end of the finished row, and the books on the lower shelf it may need to slide toward the left arm.
8. The left arm reaches every stack and every non-book on the lower shelf, the hand-over place, and the bin, the bookend, and the bare part of the side table, without extending to a joint limit.
9. Each arm reaches its own half of the working shelf board and lip, its own end of the lower shelf for the cloth, and the hand-over place. The right arm also reaches the middle of the lower shelf, for the cloth in Config M.
10. Each arm enters and leaves both shelf openings **from the front and level**. Neither arm can come down onto a shelf board from above. Neither wrist nor forearm touches the shelf above or a lip on the way in or out.
11. The right arm carries the tallest book upright into the working shelf opening, level, without the book or the wrist touching the overhead shelf.
12. If a zone cannot be reached, re-park the base by the Base positioning steps until lines 7 to 11 hold.

Materials checklist

1. The **working shelf** is one shelf of the bookcase with a **shelf directly above it**. So the way in is the front-facing **shelf opening** and never the top.
2. The shelf opening is tall enough that the tallest book, held upright by a gripper, goes in level and is set down on the board with clear space above it.
3. The working shelf board is level and **bare**. Nothing stands on it at episode start. It carries **visible dust on the board and along the shelf lip** from end to end. The dust shows on camera.
4. The working shelf board carries no labels, marks, tape, dividers, or slots. You judge every place on it by eye against the end panels, the shelf lip, and the books already placed.
5. Both **end panels** are fixed. Each stands taller than the tallest book. The row starts at the one on the right.
6. The working shelf is long enough to hold the whole finished row plus the bookend, with clear board still left over at the open end.
7. The **lower shelf** is the shelf directly below the working shelf. The working shelf board above it makes it a front-approach shelf too. Its opening is tall enough for a gripper to enter level and close on a flat book.
8. The **loose books** lie on the lower shelf in **stacks**. Some may lie too far right for the left arm to reach; the right gripper slides those toward it in Step 3.2. No stack touches another. There is clear board beside each stack, toward the middle, a little longer than a book. No stack covers the middle of the lower shelf, so the cloth rest in Config M stays clear.
9. Every loose book **lies flat on a cover**. Its **spine points toward the front of the shelf**. Its short edges are to the left and right. Which way up its title faces does not matter. No loose book stands, leans, lies spine up, or lies with its spine toward the back.
10. **Every stack is in height order, tallest on top**. This holds for every stack, with no exception. So the tallest book still to be shelved is always one that is already lying free on top.
11. The **books are all visibly different in height**. Each is clearly longer than the next as they lie, so which one is longest is never a close call. Their lengths show clearly on camera from the front.
12. No loose book overhangs the lower lip. No loose book is pushed hard against the back of the lower shelf.
13. Every book is stiff enough that it does not sag or flop when held by its covers. It is light enough for one gripper to hold by its covers, and thick enough for a gripper to close on its two covers without crushing it.
14. Every book is short enough front to back to stand fully on the working shelf board with its page edge clear of the shelf back.

15. The **non-books** sit on the lower shelf among the stacks, all on the open side of the shelf midline. No two touch. Each has a gripper's width of clear space beside it.
16. Each non-book sits steady on the bare board on its own. It is closed and dry. One gripper lifts it without its contents spilling or its parts coming apart.
17. The **cloth rest** is on the lower shelf at the place for the config (left end in Config L, middle in Config M, right end in Config R), clear of the stacks. The folded cloth lies on it flat and damp. The rest holds nothing else.
18. The **side table** is a small table standing beside the bookcase on the left, its top about level with the lower shelf and its front about level with the front of the bookcase. It holds the **bin** and the **bookend**, with a bare part left for the cloth. It is never moved or leaned on.
19. The bin stands open on the side table. Its mouth is clear and faces up. It is empty. It is deep enough for every non-book. It does not tip when one is released into it.
20. The **bookend** is a **solid block** with two flat parallel faces and its own flat base. It has no foot that slides under the books. It stands upright unaided. It does not rock. It is heavy enough to hold the whole row in place on its own.
21. The bookend stands upright on its own base on the side table, with its flat faces facing left and right. That is the same way it will stand on the working shelf, so it never has to be turned.
22. The side table has clear space above the bin, the bookend, and the bare part, reachable straight in from the front.

Workspace layout

Nothing anywhere is marked or taped out. You judge every place below by eye against the bookcase itself: the shelf lips, the end panels, and the objects already in place.

- **Bookcase:** the unit the base is parked at. It is never moved, never leaned on, and never pushed.
- **Working shelf:** the shelf being tidied. The books are shelved on it. It starts bare.
- **Overhead shelf:** the shelf directly above the working shelf. Nothing ever touches it.
- **Shelf opening:** the front-facing gap between the working shelf board and the overhead shelf. Every arm goes in and comes out through it, level and from the front.
- **Shelf board:** the flat surface of the working shelf. It is wiped before any book is placed on it.
- **Shelf lip:** the narrow front edge face of the working shelf board. It runs the full length and faces the front. It is wiped along with the shelf board.
- **Shelf back:** the panel behind the working shelf. Books stand clear of it.
- **Row end panel:** the fixed right end panel of the working shelf, where the row starts.
- **Open side:** the left end. The stacks, the non-books, and the side table are on this side. The bookend ends here.

- **Shelf midline:** the point halfway along the shelves, judged by eye. It splits the shelves into a **left half** and a **right half**, one for each gripper.
- **Row:** the books standing on the working shelf board. It starts at the row end panel end and grows toward the open side. The tallest book is at the row end panel end. The shortest is at the open end.
- **Build place:** the stretch of clear working shelf board immediately on the open side of the row, at the front of the board. Every book is set down here. It moves toward the open side as the row grows.
- **Lower shelf:** the shelf directly below the working shelf. The stacks, the non-books, and the cloth rest start on it. It is bare at episode end.
- **Lower lip:** the front edge of the lower shelf board. Books are pushed forward over it to be gripped. It is not wiped.
- **Stacks:** the piles of flat books on the lower shelf. They are gone by the end of the episode.
- **Cloth rest:** the place on the lower shelf (left end, middle, or right end) where the folded cloth lies at episode start.
- **Side table:** the small table beside the bookcase on the open side. It holds the bin, the bookend at episode start, and the used cloth at episode end.
- **Bin:** the open bin on the side table. Empty at episode start. Holds the non-books at episode end.
- **Hand-over place:** in front of the shelf lip at the shelf midline, outside the shelf opening. The only place anything changes grippers. The cloth and every book change grippers here.

Arm lanes

The two arms never cross. This holds for the whole episode.

- The left arm works the left. The right arm works the right.
- The **left gripper always stays left of the right gripper**. When both are on a shelf at once, the left gripper is nearer the left end and the right gripper is nearer the right end.
- Neither arm reaches over, under, around, or past the other.
- Neither arm reaches across the front of the other arm's body.
- Only the cloth, the books, and the bookend pass between the grippers. They pass only at the hand-over place, by a **hand over**.

Arm assignments

- **Left gripper:** clears the non-books into the bin, picks each book off the lower shelf and hands it over, lays the used cloth on the side table, and hands the bookend over. It never reaches the right end panel.
- **Right gripper:** slides a book on the lower shelf when needed so the left gripper can take it, takes each book, stands it up, shelves it, and presses its spine back. It takes the bookend

and sets it at the end of the row. It never lifts a book off the lower shelf and never touches the side table or a non-book.

- **Cloth:** the **left gripper** takes the cloth off the rest in Config L, the **right gripper** in Config M and Config R. Each gripper wipes the half of the board and lip on its own side.
- Only one arm moves an object at a time, except during a hand over.

Vocabulary

- **Book:** a bound stack of pages with a front cover, a back cover, and a spine.
- **Non-book (observable list):** anything on the lower shelf that is not a book or the cloth, such as a mug, a small picture frame, or a coiled cable. No book is ever treated as one. This list removes judgment calls and is **unvalidated**.
- **Spine:** the bound back of the book, the narrow face that carries the title. It is the face that must end up looking out the front of the shelf.
- **Page edge:** the open edge of the book, opposite the spine.
- **Bottom edge:** the short edge a book stands on when it is shelved. **Top edge** is the edge opposite it. Either short edge may become the bottom edge: which way up the title faces does not matter. When the right gripper holds a book by its right end, the bottom edge is the left end.
- **Covers:** the two large flat faces of the book. A gripper closes on these two faces. It never closes on the pages, the spine, the page edge, or a corner.
- **Free book:** a book with nothing lying on top of it. It is either the top book of a stack or a book lying alone on the lower shelf. Only free books are worked. A book with another book on it is never dug out.
- **Longest free book:** the free book that measures longest end to end as it lies. A flat book's end to end length is its height when it stands. So the longest free book is the tallest one available. The lengths are visibly different, so there is never a tie.
- **Overhang the lip:** the book's spine edge sticks out past the lower lip far enough that a gripper finger passes under it in free air. This is what makes a flat book grippable.
- **Hand over:** passing the cloth, a book, or the bookend from one gripper to the other at the hand-over place. The giving gripper holds still. The taking gripper closes on it beside the giving gripper. Hold both for half a second. Then the giving gripper opens and moves clear.
- **Wipe a half:** with the cloth folded and flat, take it into the shelf opening level and press it on the board near the shelf back at the shelf midline. Drag it out to that half's end panel and back, then move it a cloth width toward the front and go again, in strips, until the whole half of the board is wiped. Draw the cloth straight out, level. Press it flat on the shelf lip at the midline, drag it along the lip to that end of the shelf and back, and do that lip pass twice.
- **Stand up:** turning a held flat book upright in front of the shelf. The gripper holding the book near its top edge swings the bottom edge down until the book hangs upright. The spine

keeps facing the front the whole time. The book is not turned any other way.

- **Front approach:** the gripper enters and leaves a shelf opening level and from the front. It never comes down onto a shelf board from above. Every action on either shelf is a front approach.
- **Spine out:** the book stands on its bottom edge with its spine facing the front of the shelf, so the title faces the front. A book showing a cover at the front, or showing its page edge, is not spine out.
- **Height order:** each book in the row is shorter than the one on its row end panel side. The tallest is at the row end panel end. The shortest is at the open end.
- **Snug:** every book in the row touches its neighbour cover to cover. No gap shows between any two books.
- **Set back off the lip:** the book's spine sits just inside the shelf lip. It does not stick out past it.
- **Wiped:** no dust is visible anywhere on the shelf board or along the shelf lip, end to end.

Steps

Step 1: Wipe the shelf board and lip

Goal: no dust is visible on the working shelf board or along the shelf lip. The used cloth lies on the side table.

Each gripper wipes the half of the board and lip on its own side of the shelf midline, by **wipe a half**. Neither gripper crosses the midline. Keep the cloth folded and flat the whole time. Do not let it unfold, trail, or catch on a shelf edge. Do not touch the overhead shelf. Do not lean on the bookcase.

1.1 Take the cloth and wipe the first half

- **IF Config R or Config M (cloth on the right or in the middle):** with the **right gripper**, close on the folded cloth on the cloth rest and draw it straight out of the lower shelf, level. With the **right gripper**, wipe the right half.
- **IF Config L (cloth on the left):** with the **left gripper**, close on the folded cloth on the cloth rest and draw it straight out of the lower shelf, level. With the **left gripper**, wipe the left half.

Check: no dust is visible on the first half of the board or the lip. If dust remains, wipe that part again once with the same gripper before the hand-over.

1.2 Hand the cloth over and wipe the second half

- **IF Config R or Config M (right gripper started):** hold the cloth still with the **right gripper** at the hand-over place. With the **left gripper**, close on the cloth beside it. Hold both for half a second, then open the **right gripper** and move it clear. With the **left gripper**, wipe the left half.

- **IF Config L (left gripper started):** hold the cloth still with the **left gripper** at the hand-over place. With the **right gripper**, close on the cloth beside it. Hold both for half a second, then open the **left gripper** and move it clear. With the **right gripper**, wipe the right half.

Check: the board and the lip look clean from end to end. If dust remains on the second half, wipe that part again once with the gripper now holding the cloth.

1.3 Lay the cloth on the side table

- **IF Config R or Config M:** the **left gripper** already holds the cloth. With the **left gripper**, carry the cloth to the side table and lay it flat on the bare part.
- **IF Config L:** the **right gripper** holds the cloth. Hold it still with the **right gripper** at the hand-over place. With the **left gripper**, close on the cloth beside it, hold both for half a second, then open the **right gripper**. With the **left gripper**, carry the cloth to the side table and lay it flat on the bare part.
- With the **left gripper**, do not carry the cloth over the bin or the bookend. Do not drop it into the bin. Open the **left gripper** and move clear.

Check: the cloth lies flat and folded on the side table. The cloth rest is empty.

Expected state: clean board and lip, stacks and non-books still on the lower shelf as found, cloth on the side table.

Step 2: Clear the non-books into the bin

Goal: the lower shelf holds nothing but the stacks of books. Every non-book is in the bin.

The non-books and the side table are on the left. Work one non-book at a time, the one nearest the lower lip first. Never lift two together.

The **left gripper** does this whole step. The **right gripper** waits clear of the shelves.

- With the **left gripper**, enter the lower shelf from the front. Close on the non-book at its widest visible point, only hard enough to hold it without dropping it.
- With the **left gripper**, draw it straight out of the lower shelf toward the front, level. Do not lift it toward the working shelf board. Do not drag it over a stack.
- With the **left gripper**, carry it to the bin on the side table. Do not carry it over the bookend or the cloth.
- Hold it over the bin mouth and open the **left gripper** so it drops in. Do not throw it. Do not let go outside the bin mouth.
- If a non-book misses the bin, pick it up with the **left gripper** and bin it again, once.
- Repeat until no non-book is left on the lower shelf.

Check: both grippers are clear of the shelves. The lower shelf holds nothing but the stacks. Every non-book is in the bin. If a non-book is still on the lower shelf, bin it now before Step 3.

Expected state: stacks of flat books on the lower shelf, clean board and lip, non-books in the bin, cloth on the side table.

Step 3: Shelf the books, tallest first

Goal: all the books stand in one row on the working shelf. All are spine out and in height order. The tallest is at the row end panel end of the row.

Run Steps 3.1 to 3.4 once per book, until no book is left on the lower shelf. Through this whole step the **left gripper stays left of the right gripper**.

3.1 Choose the longest free book

- Look at the books as they lie on the lower shelf. Pick out the **longest free book**. That is the longest one with nothing on top of it.
- Every stack is tallest on top. So the longest free book is always the tallest book still to be shelved.
- **Never move a book to get at the one under it.** The book underneath becomes free on its own once the book above it has gone.

3.2 Pick the book off the lower shelf

- If necessary, for example when the book lies too far right for the **left gripper** to reach without extending, bring the **right gripper**, closed, against the book's right end and slide it to the left along the lower shelf, off its stack if it is on one, until the left gripper can take it. Never lift it with the right gripper. Then move the **right gripper** clear.
- If the book is the top book of a stack within the left gripper's reach, bring the **left gripper**, closed, against its **left end**. Push it to the right along the stack until it slides off the book beneath it and settles flat on the clear board beside the stack.
- Only if its spine does not already overhang the lower lip a little: with the **left gripper** closed on the book's back edge, push the book **forward** until its spine edge **overhangs the lower lip**, far enough that a gripper finger passes under it, and no further.
- With the **left gripper**, close on the book's two covers on the overhanging spine strip, **near its left end**.
- With the **left gripper**, lift the book just clear of the board and draw it straight out of the lower shelf, level, flat, spine to the front.

With the **left gripper**, close on the covers only, never the pages, the page edge, or a corner. Do not squeeze hard enough to bend the covers. Do not lift the book toward the working shelf board or carry it over a stack. Do not stretch the left arm for a book out of its reach.

Check: the **left gripper** holds the book flat by its covers in front of the lower shelf, spine to the front. If its spine is toward the back, set it back flat on the lower shelf with the **left gripper** and leave it for the station. Do not hand over a book that will stand spine in.

3.3 Hand the book over

- With the **left gripper**, carry the book, flat, up to the hand-over place and hold it still.
- With the **right gripper**, close on the covers on the spine strip **near the book's right end**.
- Hold both grippers for half a second.
- Open the **left gripper** and move it back to the left.

Check: the **right gripper** holds the book alone, flat, spine to the front. If the book slipped or turned in the hand over, give it back the same way and repeat this substep once.

3.4 Stand the book up and shelve it

- With the **right gripper**, **stand the book up** in front of the working shelf: swing the bottom edge down until the book hangs upright, spine to the front.
- With the **right gripper**, carry the book straight into the shelf opening, level, at the build place. Do not carry it over a standing book. Do not touch the overhead shelf.
- With the **right gripper**, set the bottom edge down on the board at the build place, close beside the last book placed. The first book goes down at the **row end panel** end of the shelf.
- Only if a gap shows between it and the last book: slide it along the board with the **right gripper** until its cover meets the last book. Stop as soon as the books stop moving.
- Open the **right gripper** and draw it straight out of the shelf opening.
- With the **right gripper** closed, come in from the front. Press the new book's spine straight back until it sits just inside the shelf lip, level with the spines beside it. Then draw the **right gripper** straight out.

Check after each book: the book stands on its bottom edge, spine out. Its top edge is lower than the top edge of the book on its row end panel side. No gap shows between it and the last book. No spine sticks out past the lip.

If a gap shows, slide it again with the **right gripper**, closed, against its open-side cover. If it stands spine in or out of height order, close the **right gripper** on its covers, draw it straight out, and hand it back at the hand-over place to the **left gripper**. With the **left gripper**, lay it flat on the clear board of the lower shelf, spine to the front. Then carry on from Step 3.1.

Expected state after the last book: every book stands in one row from the row end panel end toward the open side. Their top edges step down away from the panel. All spines face out. All spines sit level just inside the lip. The lower shelf holds no book.

Step 4: Set the bookend

Goal: the bookend stands on the working shelf at the open end of the row, against the last book. It holds the row in place.

The bookend goes at the left end of the row. The left arm cannot reach there on the shelf, so the **left gripper** takes the bookend off the side table and hands it over, and the **right gripper** sets it and pushes it home.

4.1 Bring the bookend over

- With the **left gripper**, close on the bookend's two flat faces on the side table. Lift it clear of the table.
- With the **left gripper**, keep it upright on its own base with its flat faces facing left and right. Do not turn it.
- With the **left gripper**, carry it to the hand-over place and hold it still.
- With the **right gripper**, close on the bookend beside the left gripper. Hold both for half a second.
- Open the **left gripper** and move it back to the left, clear of the shelf.

4.2 Set the bookend

- With the **right gripper**, take the bookend in from the front, level, still upright. Do not pass it over a book.
- With the **right gripper**, set it down on the shelf board just on the open side of the last book. Stand it on its own base, close beside the book but not touching it.
- Open the **right gripper** and draw it straight out of the shelf opening.

4.3 Push the bookend home

- Bring the **right gripper**, closed, in from the front. Set it against the bookend's **left face, low on the face**, so the push does not tip the block.
- With the **right gripper**, push the bookend along the board toward the row until it meets the last book's cover.
- Push only until the books stop moving. Do not keep pushing. Do not push hard enough to move the bookcase.
- Draw the **right gripper** straight out of the shelf opening.

Check: the bookend stands on its own base against the last book. No gap shows between any two books. If a gap shows, bring the **right gripper** back low against the bookend's left face and push once more.

Expected state: clean board and lip. All the books in one row, tallest at the row end panel end and shortest at the bookend. All spines out and level just inside the lip. Bookend at the open end. Lower shelf bare. Non-books in the bin. Cloth on the side table.

Step 5: End the episode

- Look over the shelf once: every book stands spine out in height order with the bookend against the last book, the board and lip are clean, the non-books are in the bin, and the cloth

is on the side table. Fix anything that is not.

- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Pick the next episode's config.
2. Take the bookend off the working shelf. Stand it upright on the side table, on its own base, flat faces facing left and right.
3. Take the books off the row. Lay them back on the lower shelf in **stacks**. No stack touches another. Leave clear board beside each stack, toward the middle.
4. Lay every book flat on a cover, with its **spine toward the front**. No book overhangs the lower lip.
5. Build every stack in **height order, tallest on top**. Check the stacks between them use every book.
6. Take the non-books out of the bin. Set them on the lower shelf among the stacks, all of them on the open side of the shelf midline. None touches another. Each has a gripper's width of clear space beside it. Leave the bin empty on the side table.
7. Put dust back on the shelf board and along the shelf lip from end to end, so the next episode has something to wipe off. Check it shows on camera. Leave the working shelf board bare.
8. Take the cloth off the side table. Rinse or replace it. Fold it flat and damp and lay it on the cloth rest for the config: left end, middle, or right end of the lower shelf.
9. Replace a book that sags or flops when held by its covers, or whose covers a correct grip will no longer hold.
10. Replace a bookend that rocks on its base, is too light to hold the whole row, or slides along the board under a correct low push instead of moving the row.
11. Replace a non-book that comes apart or will no longer sit steady on the bare board.
12. Clear any debris off both shelves and the side table. Check the bookcase is still steady and both end panels still fixed.
13. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the config: the violations below hold for both configs. Only the cloth start moves between them. Anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the bookcase: the gripper away from the cloth takes it off the rest; the cloth is not handed back to the left gripper in Config L; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the cloth rule; the IF line followed in Step 1 is the one for the config on the bookcase.
- **Coaching note:** look where the cloth is before the first move, then follow that IF line.

Violation: Base moved during the episode

- **Visible cue:** the bookcase shifts in frame, the shelf lip changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 5. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Arms crossed

- **Visible cue:** the left gripper goes to the right of the right gripper, or the right gripper goes to the left of the left gripper. One arm reaches over, under, around, or past the other. One arm reaches across the front of the other arm's body. A gripper carries the cloth past the shelf midline. The right gripper goes to the side table. The left gripper reaches the right end panel.
- **SOP rule broken:** Arm lanes and Arm assignments. The left arm works the left, the right arm works the right, and the left gripper always stays left of the right gripper.
- **Coaching note:** each arm stays in its own lane. If you have to reach past the other arm, the wrong arm is doing the work.

Violation: Approached from above or touched the shelf above

- **Visible cue:** a gripper comes down onto a shelf board from above instead of entering level from the front. Or a wrist, a forearm, or a held book knocks, scrapes, or leans on the shelf above or a lip on the way in or out.
- **SOP rule broken:** Steps 1 to 4. Every action on either shelf is a front approach, level, in and out through the shelf opening.
- **Coaching note:** straight in from the front, straight out the same way. There is no way in from the top.

Violation: Book carried wrong

- **Visible cue:** a book is lifted toward the shelf above it, carried over a stack or a standing book, or carried in or out of a shelf opening tilted instead of level. Or it is set down anywhere other than the build place.
- **SOP rule broken:** Steps 3.2 to 3.4. A book is drawn straight out of the lower shelf level, carried flat to the hand-over place, and carried straight into the working shelf upright to the build place.
- **Coaching note:** out level, up in front of the shelf, in level. Never over another book.

Violation: Wrong order

- **Visible cue:** anything on either shelf or the side table is touched before the shelf board and lip are wiped and the cloth is on the side table. A book is picked before every non-book is in the bin. The bookend goes onto the working shelf before the last book is placed. Or a step is skipped.
- **SOP rule broken:** Steps 1 to 4. Wipe the board and lip and put the cloth on the side table, clear the non-books, shelve the books, then set the bookend.
- **Coaching note:** wipe, clear, shelve, bookend.

Violation: Non-book left on the shelf or a book binned

- **Visible cue:** a non-book is still on the lower shelf when the first book is picked. Or a book is carried to the bin.
- **SOP rule broken:** Step 2. Clear every non-book off the lower shelf into the bin and bin nothing else.
- **Coaching note:** the non-books go in the bin and nothing else does. Look over the whole lower shelf again before moving on.

Violation: Non-book thrown or released outside the bin

- **Visible cue:** the **left gripper** tosses a non-book. It lets go of it away from the bin mouth so it lands on the side table, the lower shelf, or the floor. It drags it over a stack. It carries it over the bookend or the cloth. Or it leaves it where it landed after a miss.

- **SOP rule broken:** Step 2. Draw each non-book out level, carry it clear of the other things, hold it over the bin mouth, let go into the bin, and pick up a miss once.
- **Coaching note:** over the mouth before letting go, and always pick up a miss.

Violation: Cloth handled or passed wrong

- **Visible cue:** the cloth unfolds, trails, or catches on a shelf edge. It is dropped. It changes grippers anywhere but the hand-over place, or the giving gripper opens before the taking gripper has closed. The **left gripper** takes it off the rest. The **right gripper** lays it on the side table. It ends anywhere but flat on the side table, or it is dropped into the bin.
- **SOP rule broken:** Steps 1.1 to 1.3. The cloth stays folded and flat, changes grippers only at the hand-over place, and is laid on the side table by the **left gripper**.
- **Coaching note:** folded and flat, hold still to give, close before the other opens, and it ends on the side table.

Violation: Shelf not wiped as prescribed

- **Visible cue:** dust is still visible on the board or the lip when the cloth is laid on the side table. A half of the board or the lip is skipped. A half of the lip gets one pass instead of two. The cloth lifts off partway along a stroke. A stroke stops short of the end panel or the end of the shelf. Or the cloth comes down onto the board from above instead of entering level.
- **SOP rule broken:** Steps 1.1 and 1.2. Wipe each half of the board in strips from back to front, then run each half of the lip out to the end and back, twice.
- **Coaching note:** board first, back to front, then the lip twice. Check it is clean before the cloth goes down.

Violation: Dug under a stack

- **Visible cue:** a gripper moves, slides, or lifts a book that has another book lying on top of it. Or a book is pulled out from under a stack.
- **SOP rule broken:** Step 3.1. Work only free books, and let the book underneath become free on its own.
- **Coaching note:** never dig. The stacks are built tallest on top, so the free book is always the one you want.

Violation: Book taken out of height order

- **Visible cue:** the **left gripper** picks a book that is visibly shorter as it lies than another free book still on the lower shelf. Or the finished row has a book taller than the one to its right.
- **SOP rule broken:** Steps 3.1 and 3.4. Take the longest free book each time, so the row runs tallest at the right end panel end to shortest at the left end.
- **Coaching note:** look at all the free books before reaching in. Longest free book, every time.

Violation: Book picked wrong

- **Visible cue:** the **left gripper** lifts the top book straight off its stack instead of sliding it down onto the board first. It closes on a book whose spine does not overhang the lower lip. It grips the book near its right end instead of its left end. It stretches for a book out of its reach instead of the right gripper sliding the book near first. Or it pushes with an open gripper or a wrist.
- **SOP rule broken:** Step 3.2. A far book is slid near by the right gripper first. Slide the book down onto the clear board, push it forward only if the spine does not already overhang the lower lip, then close on the covers at the overhanging spine strip near the left end.
- **Coaching note:** too far, slide it near with the right first. A finger must fit under before you grip.

Violation: Wrong grasp of a book

- **Visible cue:** a gripper closes on a book's pages, page edge, spine alone, or a corner instead of its two covers. It holds only one cover. It squeezes hard enough to bend or crease the covers. Or the book slips, swings, or turns in the gripper while it is carried.
- **SOP rule broken:** Steps 3.2 to 3.4. Every book is held by its two covers, on the spine strip, firmly enough that it does not move in the gripper and lightly enough that the covers do not bend.
- **Coaching note:** two covers, on the spine strip, firm but not crushing.

Violation: Book handed over wrong

- **Visible cue:** the book changes grippers anywhere but the hand-over place, or inside a shelf opening. The giving gripper moves during the hand over. The **left gripper** opens before the **right gripper** has closed, or with no half-second hold. The right gripper takes it at the left end. Or the left gripper shelves the book itself.
- **SOP rule broken:** Step 3.3. Carry the book flat to the hand-over place, hold still, let the right gripper close near the right end, hold both, then open the left gripper.
- **Coaching note:** giver holds still, taker closes, half a second, then let go.

Violation: Book stood up wrong

- **Visible cue:** the **right gripper** turns the book so its spine faces back or up, rolls it, or turns it end for end. It stands the book up inside the shelf opening instead of in front of it. It drops the book onto the board instead of setting it down. Or it lets go before the bottom edge is on the board.
- **SOP rule broken:** Step 3.4. Stand the book up in front of the shelf by swinging the bottom edge down, carry it in upright, and set the bottom edge down at the build place beside the last book.
- **Coaching note:** stand it up outside, carry it in upright, set it down beside the last book.

Violation: Book shelved spine in or face out

- **Visible cue:** a book in the row shows a cover or its page edge at the front instead of its spine.
- **SOP rule broken:** Steps 3.2 to 3.4. Pick the book spine forward and stand it up by swinging the bottom edge down so it stands spine out. Which way up the title faces does not matter.
- **Coaching note:** check the spine faces front before the pick. A book that lies spine back stands spine in.

Violation: Row not slid home or shoved

- **Visible cue:** a gap shows between two books in the row and the book is never slid to close it. Or the **right gripper** slides hard enough to bow a cover or skew the row, or keeps sliding after the books have stopped moving.
- **SOP rule broken:** Step 3.4. Set the book close beside the last book. If a gap shows, slide it until it meets the last book, and stop as soon as the books stop moving.
- **Coaching note:** slide until it meets, stop when the row stops.

Violation: Book left leaning on another

- **Visible cue:** a book in the row is let go leaning its top against the book beside it, the bookend, or the end panel, instead of standing on its own bottom edge. Or it rests on a corner with its bottom edge off the board.
- **SOP rule broken:** Step 3.4. Set the book's bottom edge down flat on the board beside the last book so it stands on its bottom edge before the gripper lets go.
- **Coaching note:** bottom edge flat on the board before you open the gripper.

Violation: Spine left sticking out past the lip

- **Visible cue:** a book in the finished row has its spine standing out past the shelf lip, or its spine sits visibly behind the spines beside it, and the **right gripper** never presses it back.
- **SOP rule broken:** Step 3.4. Press the new book's spine straight back with the closed **right gripper** until it sits just inside the shelf lip, level with the spines beside it.
- **Coaching note:** every book gets the back-press. That is what lines the fronts up.

Violation: Bookend set wrong

- **Visible cue:** the bookend lies on a face instead of standing on its base. Its flat faces end up facing front and back instead of left and right. It is turned on the way. It is set on top of a book. The **left gripper** reaches onto the working shelf to set it instead of handing it over. Or it is set somewhere other than at the left end of the row.
- **SOP rule broken:** Steps 4.1 and 4.2. The **left gripper** takes the bookend off the side table upright and hands it over. The **right gripper** sets it on its own base at the left end of the row, flat faces facing left and right.
- **Coaching note:** left brings it, right sets it, the same way it stood on the side table.

Violation: Bookend not pushed home, or pushed too far

- **Visible cue:** the bookend is left standing clear of the last book. It is pushed while still gripped instead of pushed by the closed **right gripper**. It is pushed high on its face so it tips. Or the push carries on after the books stop and bows the row or shifts the bookcase.
- **SOP rule broken:** Step 4.3. Let go, come back low and closed on the bookend's left face, push until it meets the last book, and stop there.
- **Coaching note:** low on the face, push until the books stop, then stop.

Violation: Dropped, toppled, or knocked over

- **Visible cue:** a book, a non-book, the cloth, or the bookend is dropped. A stack is knocked over. A book already placed is toppled or pushed off the shelf on the way in or out. The row collapses. Or the bin is knocked over.
- **SOP rule broken:** Steps 1 to 4. Nothing is dropped, toppled, or knocked out of its place, and every gripper leaves a shelf opening straight out to the front.
- **Coaching note:** check the path and the landing place before the arm moves. Draw straight out so a finger does not catch a spine.

Violation: Wrong arm used

- **Visible cue:** the **right gripper** touches a non-book or the bin, or lifts a book off the lower shelf. The **left gripper** sets a book or the bookend on the working shelf, touches a book already in the row, or reaches the right end panel. Or anything other than the cloth, a book, or the bookend changes grippers.
- **SOP rule broken:** Steps 1 to 4. The **left gripper** works the non-books, picks the books, and hands the bookend over. The **right gripper** slides a book near when needed, stands the books up, shelves them, and sets the bookend. Only the cloth, the books, and the bookend change grippers, at the hand-over place.
- **Coaching note:** left picks, right shelves.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a non-book or a book on the lower shelf, dust on the board or lip, a book spine in or out of height order, a gap in the row, a spine sticking out past the lip, the bookend missing or clear of the row, the cloth not on the side table, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 5. Look over the shelf and fix anything that is not right, then return both arms home with grippers open and stop recording.
- **Coaching note:** look over the shelf first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot show the shelf board, the lip, or the book lengths clearly**, so whether the shelf is clean or whether the row is in height order cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Shelf opening too tight:** an opening so low that the tallest book cannot be carried in upright without touching the overhead shelf, or a lower shelf a gripper cannot enter level from the front.
- **Bookcase not steady:** a unit that rocks, walks, or leans when a book is pushed along a board or the bookend is pushed home.
- **Shelf board will not hold a book:** a board that is not level, or so slick that a correctly placed book will not stay where it is put, or so rough that a correct push will not slide a flat book along it.
- **Lower lip too shallow to grip past:** a lip that will not let a book's spine overhang far enough for a gripper finger to pass under it.
- **Defective book:** one that sags or flops when held correctly by its covers, that is too heavy to hold by its covers, or whose covers slip out of a correct grip.
- **Two books too close in length to tell apart**, so which free book is longest cannot be seen from the front.
- **Defective bookend:** one that rocks on its base, will not stand upright unaided, is too light to hold the whole row, or slides along the board under a correct low push instead of moving the row.
- **Defective non-book:** one that comes apart, spills, or will not sit steady on the bare board.
- **Defective cloth:** one that is torn, shedding, dry at episode start, or so loaded with dust that a correct pass only smears the shelf.
- **Damaged lip surface:** a stain, scratch, or mark in the shelf lip itself that does not lift under a correct pass.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the side table, the cloth rest, a stack, the hand-over place, the build place, or the row end panel cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Take the cloth from the cloth rest
2. Wipe the first half of the shelf board and lip
3. Hand the cloth over at the shelf midline
4. Wipe the second half of the shelf board and lip
5. Lay the cloth on the side table
6. Move one non-book from the lower shelf into the bin
7. Slide a book toward the left arm with the right gripper (when needed)
8. Push the book forward so its spine overhangs the lower lip
9. Pick the book and draw it out of the lower shelf
10. Hand the book over to the right arm
11. Stand the book up in front of the working shelf
12. Set the book at the build place beside the last book
13. Press the book's spine back off the lip
14. Hand the bookend over to the right arm
15. Set the bookend at the open end of the row
16. Push the bookend home against the last book
17. Return both arms home and end the episode